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Mapping irrigable land potential for surface irrigation through GIS-based multi-criteria analysis in Ethiopia's Gidabo watershed

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Abstract

Identifying land suitable for surface irrigation is essential for improving agricultural production while ensuring efficient use of land and water resources. However, such assessments are often challenging in data-scarce regions where terrain, soil conditions, land use, and access to water and infrastructure vary widely. This study evaluated land suitability for surface irrigation in the 3180 km² Gidabo Watershed in Ethiopia's Rift Valley Lakes Basin. Seven biophysical and accessibility factors were analyzed: slope, soil depth, soil texture, soil drainage, land use/land cover, and distance to rivers and roads. These factors were standardized using FAO guidelines and local expert knowledge and integrated through the Analytical Hierarchy Process (AHP) and GIS-based weighted overlay analysis. The results indicate that slope was the most influential factor, with a weight of 46.34%, followed by distance to rivers (28.51%) and soil characteristics (14.46%). The AHP consistency ratio of 0.053 indicated acceptable consistency in the expert judgments. Overall, 93.23% of the watershed was physically suitable for surface irrigation to varying degrees (S1–S3). However, only 35.79% (113,831 ha) fell within the highly and moderately suitable classes (S1–S2), representing areas with more favorable conditions for irrigation development. Highly suitable land (S1) accounted for 4.32% of the watershed and was mainly located in valley bottoms and lowland areas along the Gidabo River and its major tributaries. In contrast, steep slopes exceeding 8%, which cover about 62% of the watershed, represent the main physical constraint. The distribution of S1–S2 land also varied considerably among sub-basins, ranging from 471 to 21,486 ha. These findings highlight that while most of the watershed has some level of physical suitability, the area with favorable conditions for practical irrigation development is much more limited. The resulting suitability map provides a useful tool for prioritizing areas for detailed feasibility studies and guiding irrigation investment planning in the Gidabo Watershed and similar data-scarce watersheds in sub-Saharan Africa.

Keywords: Surface irrigation, Land suitability, GIS-MCDA, AHP, Ethiopia, Gidabo watershed



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1 Introduction

The increasing demand for food production, driven by global population growth, presents persistent challenges in balancing agricultural productivity with sustainable resource use [3, 28]. Irrigated agriculture occupies only 20% of cultivated land but contributes nearly 40% of global food production, underscoring its critical role in stabilizing food supplies [18, 38]. Assessing land suitability for surface irrigation is therefore a strategic priority, particularly in regions vulnerable to climate variability and rapid population growth [8, 18].

Land suitability assessment (LSA) provides a scientific basis for sustainable development by identifying where irrigation can be implemented cost-effectively and with minimal environmental degradation [5, 6]. Integrating Geographic Information Systems (GIS) with Multi-Criteria Decision Analysis (MCDA), particularly the Analytical Hierarchy Process (AHP), has proven highly effective for mapping irrigation suitability [4, 12]. AHP organizes complex problems into hierarchical structures, enabling systematic decision-making through pairwise comparisons [35].

Ethiopia exemplifies both the potential and constraints of irrigation development in sub-Saharan Africa. Agriculture contributes approximately 44% of GDP and employs nearly 80% of the population, yet despite an estimated 3.7 million hectares of irrigable land, less than 15% has been developed [10, 18, 22, 31, 33]. The sector remains predominantly rain-fed, making it highly vulnerable to rainfall variability and climate change [21]. Climate projections for East Africa indicate increasing uncertainty in precipitation patterns [27]. Recognizing this, the Government of Ethiopia has prioritized irrigation through successive national development strategies, including the Growth and Transformation Plans and the current Ten-Year Development Plan (2020/21–2029/30) (EPDC [17]). These strategies recognize irrigation as central to agricultural transformation and food security, with targets to develop substantial new irrigated areas [33]. However, planning has often proceeded without detailed assessments of land suitability, leading to operational challenges, including water shortages and underutilization of infrastructure [1, 8].

Several studies have addressed this gap in Ethiopian basins. Girma et al. [24] found 8.5% of the Omo-Gibe Basin highly suitable for surface irrigation. Hagos et al. [26] reported 63% of the Minch Yekest catchment as highly suitable. Muluneh et al. [34] assessed the Rift Valley Lakes Basin (RVLB) and found that 6.2% of the land was highly suitable. More recently, Legesse et al. [29] identified slope (40.74%) as the dominant limiting factor in the upper Awash Basin, with only 35.8% of the area highly to moderately suitable. Similarly, Abagissa et al. [2] reported that 36% of the Gololcha Watershed was unsuitable due to slope, which received the highest weight (46.36%) in their AHP analysis.

Despite these efforts, a comprehensive LSA for surface irrigation in the Gidabo Watershed is still lacking. This gap is consequential because the watershed possesses three distinctive characteristics: (i) it straddles the steep eastern escarpment of the Main Ethiopian Rift (elevation 1171–3213 m), generating sharp transitions in slope and soil depth; (ii) its land-use mosaic smallholder enset-coffee-maize systems with agroforestry, differs from cereal-dominated or pastoral systems; (iii) its deeply incised river valley requires locally validated proximity-to-river thresholds. Beyond watershed-specific needs, this

study introduces three methodological innovations: (i) transparent documentation of expert panel composition and voting outcomes; (ii) a two-round Delphi-style threshold validation with explicit consensus criteria; and (iii) sub-basin dual ranking (absolute and percentage area) to distinguish sub-basins suitable for central infrastructure from those better served by distributed small-scale schemes.

This study addresses this knowledge gap by conducting a systematic GIS-based multi-criteria LSA for surface irrigation in the Gidabo Watershed. The specific objectives are to: (i) map the spatial distribution of suitability parameters; (ii) determine their relative weights using AHP with rigorous consistency assessment; (iii) integrate weighted criteria through GIS-based weighted overlay; and (iv) quantify suitable lands at watershed and sub-basin scales. In general, this study addresses the question: Where, and under what biophysical and infrastructure constraints, is surface irrigation economically viable in the Gidabo Watershed? By providing a spatially explicit, reproducible assessment, this study aims to inform the prioritization of irrigation investments and offer a transferable framework for data-scarce regions in sub-Saharan Africa.

2 Study area description

The Gidabo Watershed is situated in the Rift Valley Lakes Basin of Ethiopia, between latitudes $6^{\circ}05'$ to $6^{\circ}45'$ N and longitudes $38^{\circ}05'$ to $38^{\circ}35'$ E (Fig. 1). The watershed covers approximately 3,180.49 km². The watershed's topography is characterized by a steep

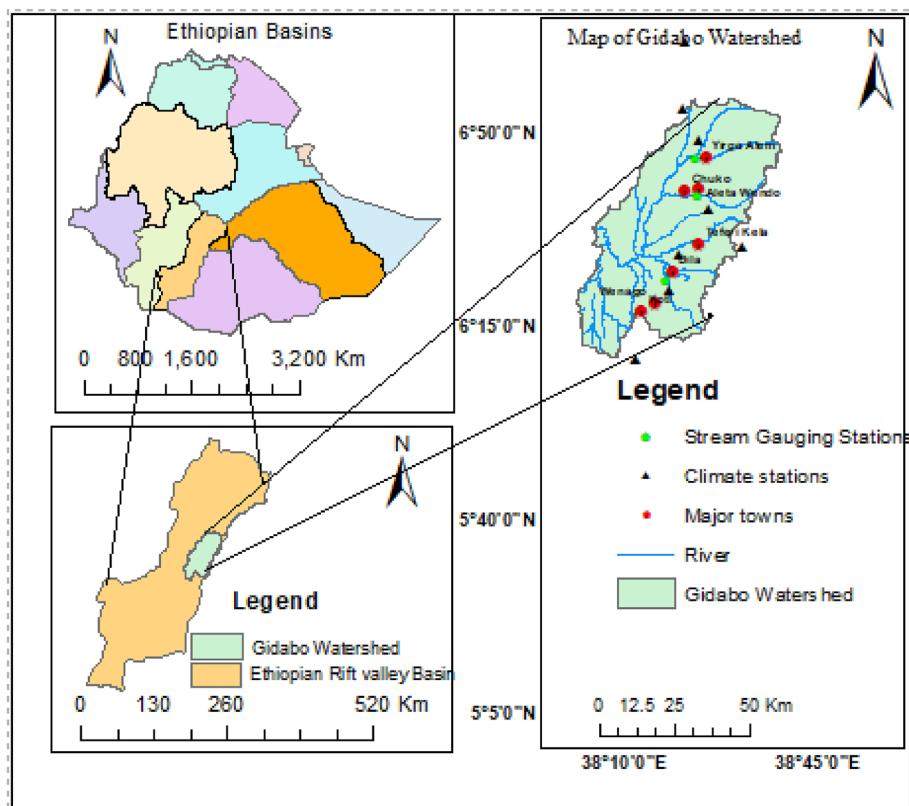


Fig. 1 Location map showing Ethiopia's major river basins, Rift Valley Lakes Basin, and Gidabo watershed with drainage network, and gauging stations

elevation gradient from 3213 to 1171 m, yielding a weighted-average slope of approximately 14.98%. The Gidabo River originates from the Soka Sonicha mountain range, flows approximately 120 km southward along the eastern floor of the Great Rift Valley, and terminates in Lake Abaya.

The climate is characterized as sub-humid, with a bimodal rainfall pattern: the main rainy season occurs from March to May (Belg), and a secondary season from September to November (Meher). Mean annual rainfall is 1,385 mm (range: 900–1800 mm), with mean annual temperature ranging from 15 °C in highlands to 25 °C in lowlands. The watershed supports diverse land uses, including agriculture (cereals, coffee, enset, vegetables), mixed forests, grasslands, and settlements.

3 Methodology

This study employed an integrated approach combining Geographic Information System (GIS) techniques with the AHP and MCDM methods to evaluate and map land suitability for surface irrigation development in the Gidabo Watershed. The methodology framework is presented in Fig. 2.

3.1 Data sources and preparation

3.1.1 Digital elevation model (DEM)

A 30 m resolution SRTM DEM was acquired from the USGS Earth Explorer platform (<http://earthexplorer.usgs.gov>). The DEM was used for watershed delineation, slope analysis, and derivation of topographic parameters.

3.1.2 Land use/land cover (LULC)

Landsat 8 OLI/TIRS imagery (path 168, rows 55–56, February 2020, 30 m resolution) was obtained from USGS. Image preprocessing, including layer stacking, geometric correction, and sub-setting, was performed in ERDAS Imagine. Supervised classification using the maximum likelihood algorithm was employed, supported by 249 ground truth points collected during field surveys, supplemented by Google Earth imagery and topographic maps.

3.1.3 Soil data

A 1:250,000-scale soil map was obtained from the Ministry of Water and Energy (MoWE) of Ethiopia. Soil types were classified according to the FAO/UNESCO soil classification system. Soil physical properties (texture, depth, and drainage) were derived from the soil map to characterize irrigation suitability.

3.1.4 Infrastructure data

Road network data were obtained from the Ethiopian Roads Authority (ERA) geodatabase (Ethio-Roads), while river network data were extracted from the Ethio-Rivers database maintained by the Ministry of Water and Energy. Road network data included all-weather primary asphalt roads, secondary gravel roads, and major regional feeder routes. Euclidean distance tools in ArcGIS were used to generate proximity maps (in meters) for both roads and perennial rivers, representing accessibility to markets and water sources, respectively (Fig. 3).

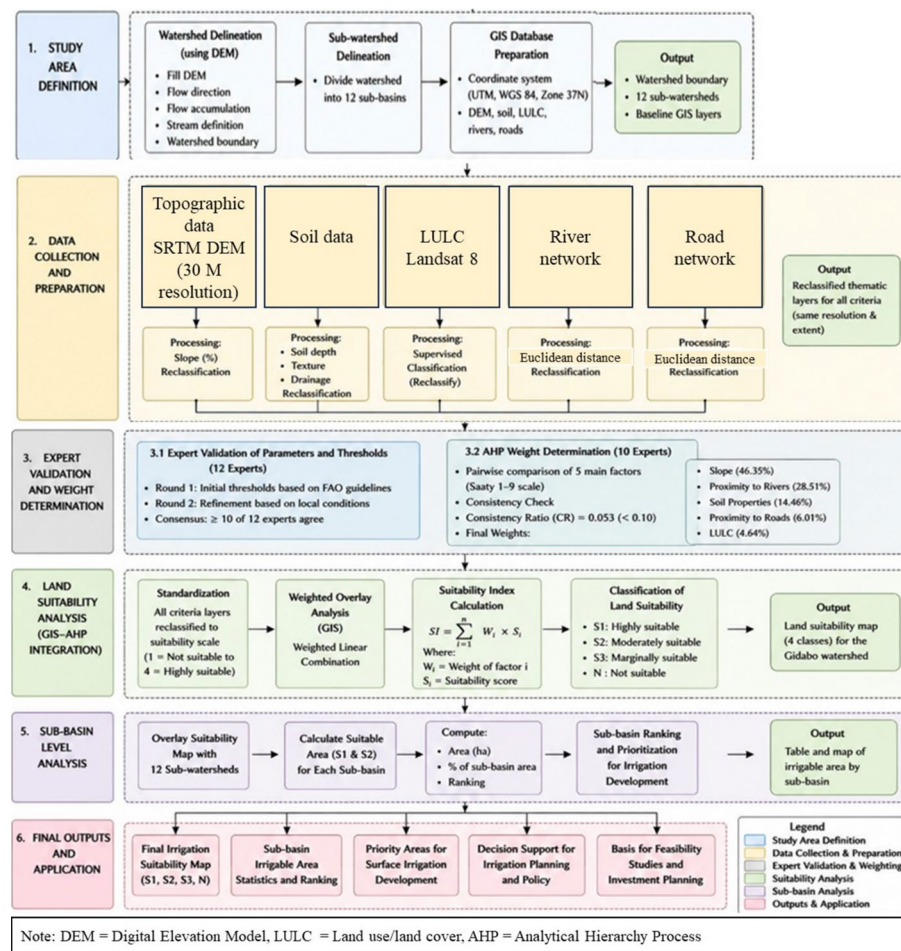


Fig. 2 Methodological framework used for surface irrigation suitability assessment in the Gidabo Watershed. The framework integrates GIS-based spatial analysis, FAO land suitability classification, expert-based Analytical Hierarchy Process (AHP) weighting, weighted overlay analysis, and sub-basin prioritization to identify potentially irrigable lands for surface irrigation development

3.2 Selection of suitability factors

Based on an extensive literature review [1, 2, 11, 25, 29], FAO guidelines [19], and consultations with local agricultural and water resources experts, seven key parameters were identified as critical determinants for surface irrigation suitability: slope, soil depth, soil texture, soil drainage, land use/land cover (LULC), proximity to rivers, and proximity to roads. Each parameter was selected based on its direct or indirect influence on the feasibility, cost-effectiveness, and sustainability of surface irrigation systems.

To ensure the credibility and relevance of the selected criteria, a panel of experts was convened to participate in two structured exercises: (i) validation of the seven selected parameters (confirmation that no critical factor was omitted), and (ii) adjustment of classification thresholds based on local conditions. The panel comprised 12 experts from three institutional categories, selected based on their disciplinary expertise, professional experience in irrigation development, and familiarity with the Gidabo Watershed. Table 1 summarizes the composition of the expert panel.

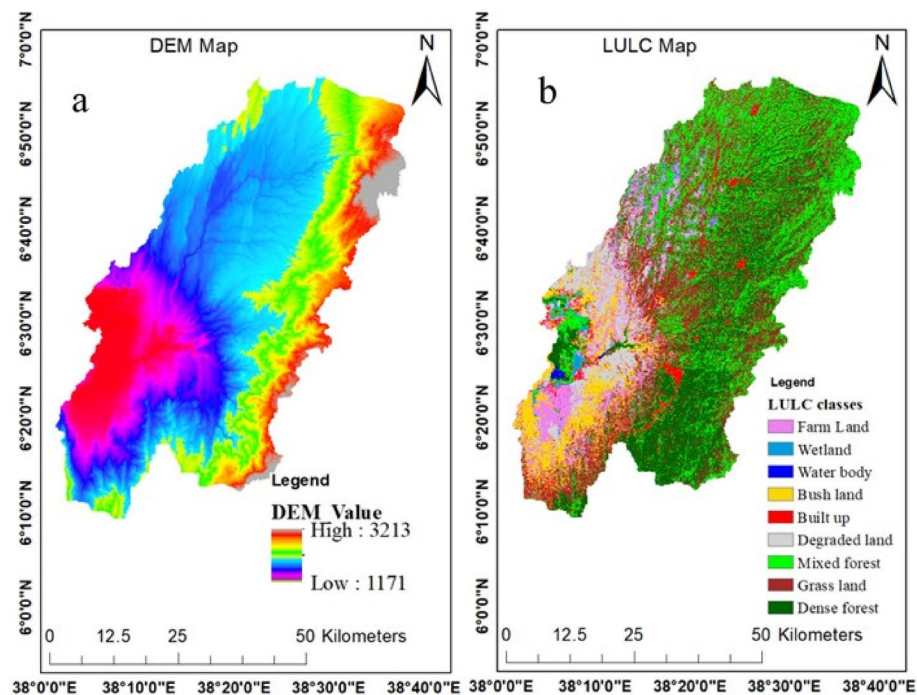


Fig. 3 Digital elevation model (a) and land use/land cover (b) maps of the study area

3.3 Criteria standardization and reclassification

All spatial datasets were projected to WGS 1984 UTM Zone 37N and resampled to a 30-m grid using the nearest-neighbor method to maintain data integrity. Euclidean distance tools in ArcGIS 10.8 were used to generate proximity maps. This ensured pixel-to-pixel compatibility for subsequent overlay analyses. Each factor was reclassified into four suitability classes according to the FAO framework: Highly Suitable (S1), Moderately Suitable (S2), Marginally Suitable (S3), and Not Suitable (N). The classification thresholds were established based on an extensive review of peer-reviewed literature [2, 3, 11–13, 25, 26, 32] and FAO guidelines, and expert knowledge of local conditions. Table 2 presents the detailed suitability classification criteria for each parameter. Thresholds were determined through a three-step process: (i) initial values were extracted from FAO guidelines and peer-reviewed studies in Ethiopian and East African basins; (ii) these values were presented to the expert panel for validation and adjustment based on local conditions; (iii) final thresholds represent consensus among at least 10 of the 12 experts following two rounds of discussion.

For slope thresholds, the 0–2% range (S1) follows FAO [19] and Abagissa et al. [2] as ideal for surface irrigation, while 2–5% (S2) is permissible with moderate terracing based on Belay and Bewket [9] and Dawit et al. [13], the 5–8% range (S3) requires significant earthworks and was validated by the expert panel based on local scheme experience, and >8% (N) was unanimously agreed as infeasible without major terracing. For soil depth, >100 cm (S1) follows FAO's "very deep" classification and local soil surveys, 50–100 cm (S2) is from Gonfa et al. [25], 25–50 cm (S3) from Legesse et al. [29], and <25 cm (N) from FAO [19] confirmed by Lithic Leptosols mapping. Soil texture thresholds follow FAO [19] with loams as S1, sandy loam and clay loam as S2 (adjusted from S3 in some

Table 1 Composition and credentials of the expert panel involved in factor selection and threshold validation

Category	Number of experts	Educational background	Professional training/ specialization	Current occupation and institution	Relevance to the study	Voting outcomes for key threshold adjustments	Specific contribution
Academic researchers (water resources)	5	PhD (4), MSc (1) in Water Resources Engineering, Hydrology, or Agricultural Water Management	Hydrological modeling, irrigation engineering, GIS-based suitability analysis	Assistant Professors, Dilla University (Department of Water Resources Engineering)	Direct research experience in the Gidabo Watershed (3 experts) or adjacent Rift Valley basins (2 experts); previously published on SWAT modeling and land suitability in the region	Slope S2 upper bound (2–5%); 11/12; Soil depth S1 (> 100 cm); 12/12; Distance to river or adjacent Rift Valley S2 (2–3 km): 10/12; Distance to road S2 (2.5–4.5 km): 11/12; Soil texture (clay loam S2): 10/12	Validated factor selection; proposed adjustments to slope and soil depth thresholds based on local conditions; reviewed classification criteria
Government water resources specialists	4	MSc (3), BSc (1) in Irrigation Engineering, Hydraulic Engineering, or Water Resources Management	Irrigation scheme design and operation, water allocation, and river basin management	Senior irrigation engineers/basin planners, Ministry of Water and Energy (MoWE) and Rift Valley Lakes Basin Authority	Direct responsibility for irrigation planning and water allocation in the Gidabo sub-basin; operational knowledge of existing small-scale irrigation schemes	Distance to river S2 (2–3 km): 10/12; Distance to road S2 (4–4.5 km): 11/12; Slope thresholds: 11/12; Soil drainage (imperfect S3): 11/12	Provided local knowledge on practical constraints (e.g., canal construction costs, water abstraction regulations); adjusted distance-to-river and distance-to-road thresholds
Agricultural experts (crop & soil)	3	MSc (2), BSc (1) in Soil Science, Agronomy, or Crop Science	Soil classification, crop water requirements, and agricultural extension	Senior agricultural officers/soil scientists, Sidama and Gedo Zone Agricultural Offices	Daily interaction with farmers in the watershed; deep understanding of local cropping patterns; soil conditions, and irrigation challenges	Soil texture (clay loam S2 vs. S3): 10/12; Soil depth S3 (25–50 cm): 10/12; LULC classification: 11/12	Validated soil parameter thresholds (depth, texture, drainage); confirmed LULC classifications; provided cropping calendar data

Table 2 Suitability classification criteria for the selected parameters

Criterion	Suitability class and thresholds				Consensus
	S1	S2	S3	N	
Slope (%)	0–2	2–5	5–8	> 8	11/12
Soil Depth (cm)	> 100	50–100	25–50	< 25	12/12
Soil Texture	Loam, Silt Loam	Sandy Loam, Clay Loam	Sand, Loamy Sand	Clay, Sandy Clay	10/12
Soil drainage	Well	Moderately well	Imperfect	Poor	11/12
Land Use/Land Cover	Agricultural land	Range grasses,	Forest-mixed, Range bush	Forest-evergreen, built-up, wetland, water bodies, degraded land	11/12
Distance to River (km)	0–2	2–3	3–4	> 4	10/12
Distance to Road (km)	1.5–2.5	2.5–4.5	4.5–5.5	> 5.5	11/12

studies based on expert panel 10/12 agreement), sand and loamy sand as S3, and clay as N. Soil drainage follows FAO [19] and Birhanu et al. [11], with imperfect drainage downgraded from S2 to S3 based on observed waterlogging in Vertisols (11/12 agreement). Land use/land cover thresholds classify agricultural land as S1, range grasses as S2, forest-mixed and range bush as S3 (adopted from Dinku and Kebede [16], and protected or unsuitable classes as N. Distance to river thresholds (0–2 km S1, 2–3 km S2, 3–4 km S3, > 4 km N) were adapted from Abagissa et al. [2], with the S2 upper bound reduced from 4 to 3 km based on local topography (10/12 agreement). Distance to road thresholds (1.5–2.5 km S1, 2.5–4.5 km S2, 4.5–5.5 km S3, > 5.5 km N) follow Gonfa et al. [25] and Legesse et al. [29], with the S2 upper bound increased from 4 to 4.5 km based on local road density (11/12 agreement). The expert validation process involved two rounds: initial threshold presentation, where experts rated each threshold as "appropriate," "needs adjustment," or "inappropriate", discussion of thresholds where ≥ 3 experts indicated need for adjustment; and final voting, where consensus was defined as ≥ 10 of 12 experts agreeing.

3.4 Analytic hierarchy process (AHP) for weight determination

3.4.1 Pairwise comparison matrix

The Analytic Hierarchy Process (AHP) was employed to determine the relative importance (weights) of each parameter in influencing land suitability for surface irrigation. A hierarchical structure was established with the goal (land suitability for surface irrigation) at the top, followed by the seven evaluation parameters at the criterion level [23].

A pairwise comparison matrix was constructed based on expert judgments from soil scientists, irrigation engineers, hydrologists, and agricultural planners from Dilla University, the Ministry of Water and Energy, and the Rift Valley Lakes Basin Authority. Experts were asked to rate the relative importance of each pair of parameters using Saaty's 1–9 fundamental scale, where 1 indicates equal importance, and 9 indicates extreme importance of one factor over the other. The geometric mean of individual expert judgments was calculated to aggregate the pairwise comparisons into a single consensus matrix.

The same 12-member expert panel described in Table 1 participated in the factor validation and threshold adjustment exercises. However, the pairwise comparison matrix for AHP weight determination was completed by a subset of 10 experts. The two agricultural extension experts (from the Sidama and Gedeo Zone Agricultural Offices) were excluded from the pairwise comparison exercise because they indicated limited familiarity with comparative judgment methods during the pre-workshop briefing. Their contributions remained valuable for factor validation and threshold adjustment, but they did not participate in the AHP weighting.

3.4.2 Normalization and weight calculation

The pairwise comparison matrix was normalized by dividing each element by the sum of its column. Parameter weights (eigenvector) were then computed as the average of the normalized values across rows. The principal eigenvalue (λ_{max}) was calculated by multiplying the pairwise comparison matrix by the weight vector and averaging the ratios of the resulting vector components to the corresponding weights.

3.4.3 Consistency assessment

To ensure the reliability and logical consistency of the expert judgments, the consistency ratio (CR) was calculated following Saaty's procedure:

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (1)$$

$$CR = \frac{CI}{RI} \quad (2)$$

where CI is the consistency index, n is the number of parameters/factors, and RI is the random index obtained from Saaty's table. A CR value of ≤ 0.10 (10%) is considered acceptable [36], indicating that the judgments are sufficiently consistent; otherwise, the pairwise comparisons should be revised.

3.5 Weighted overlay analysis and suitability mapping

After reclassifying each parameter layer into suitability classes, a weighted overlay analysis was performed in ArcGIS 10.8 using the Weighted Sum tool. Each reclassified raster was multiplied by its respective AHP-derived weight, and the weighted values were summed according to the following equation:

$$SuitabilityIndex = \sum_{i=1}^n (W_i \times R_i) \quad (3)$$

where n is the number of criteria, W_i is the normalized weight of criterion i, and R_i is the reclassified score of criterion i. The resulting continuous suitability index raster was reclassified into the four final suitability classes (S1–N) (Table 3) to produce the definitive surface irrigation suitability map for the Gidabo watershed. The area statistics for each class were then calculated to quantify the basin's irrigation potential.

Table 3 Suitability class and its description

Class	Suitability	Description
S1	Highly suitable	Land without any significant limitations
S2	Moderately suitable	Moderately severe limitations (slope 2–5%, moderate distance to rivers 2–3 km, or moderately well-drained soils), which reduce productivity or increase development costs compared to S1 lands
S3	Marginally suitable	Overall, severe limitations; only marginally justifiable
N	Not suitable	Limitations not currently overcome with existing knowledge & cost

Table 4 Slope suitability classification for surface irrigation

Slope range (%)	Area (km ²)	Area (%)	Suitability Class
0–2	168.70	5.30	Highly suitable (S1)
2–5	546.22	17.17	Moderately suitable (S2)
5–8	493.33	15.51	Marginally suitable (S3)
>8	1,972.24	62.02	Not suitable (N)
Total	3,180.49	100	

3.6 Sub-basin level analysis

The 12 delineated sub-basins (derived from the 30 m DEM, with a threshold area of 3,000 ha for stream definition) were used as spatial units to calculate the irrigable-suitable area. For each sub-basin, the total area of S1 and S2 classes was extracted using zonal statistics in ArcGIS, and the percentage of sub-basin area classified as irrigable (S1+S2) was computed. This sub-basin-level disaggregation enables targeted prioritization of irrigation investments.

4 Results

4.1 Land suitability for surface irrigation

4.1.1 Individual factor suitability

4.1.1.1 Slope suitability Slope suitability is summarized in Table 4, with physically suitable areas (0–8%) covering 37.98% of the watershed, concentrated in valley bottoms along the Gidabo River (Fig. 4). Moderately suitable areas (2–5%) occupy 546.22 km² (17.17%), while marginally suitable areas (5–8%) cover 493.33 km² (15.51%). Steep slopes (>8%), comprising 62.02% (1,972.24 km²) of the watershed, were classified as not suitable, primarily in the highland escarpments of the Soka Sonicha mountain range.

The dominance of steep slopes (>8%) across 62% of the watershed represents the most significant topographic constraint to surface irrigation development in the Gidabo Watershed. This finding is consistent with studies in other Ethiopian highland watersheds, where terrain complexity substantially limits irrigation suitability. For instance, Abagissa et al. [2] reported that 36% of the Gololcha River Watershed was unsuitable for surface irrigation due to slope constraints, while Legesse et al. [29] found that 54.38% of the Dendi District in the upper Awash Basin fell within unsuitable slope classes (>8%). Similarly, Gonfa et al. [25] identified slope as the dominant limiting factor in the Becho Plain, accounting for 46.34% of the total weight in their

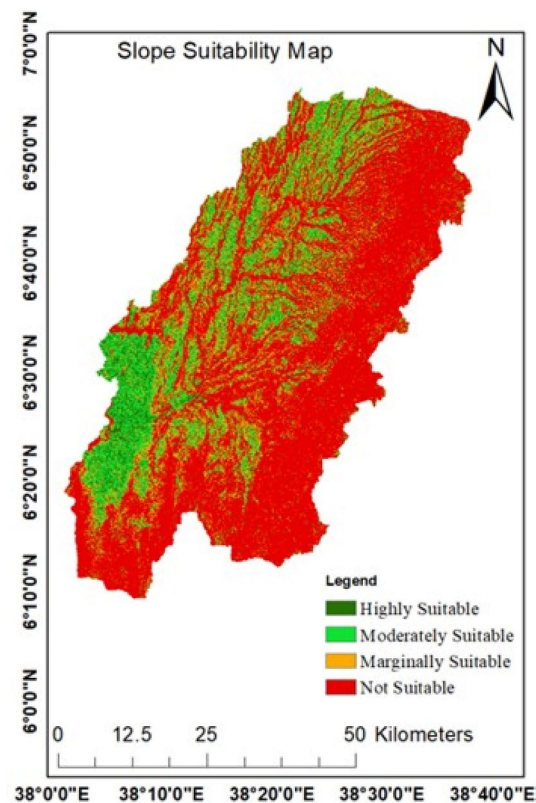


Fig. 4 Spatial distribution of slope classes and their suitability for surface irrigation

Table 5 Soil parameter suitability classes

Soil parameters		Suitability			
No		Highly suitable	Moderately suitable	Marginally suitable	Not suitable
1	Soil texture	30.67	26.43	42.89	-
2	Soil depth	50.27	16.53	7.40	25.80
3	Soil drainage	-	40.68	33.67	25.66

AHP analysis. The concentration of suitable slopes in valley bottoms and lowland plains along the Gidabo River and its tributaries aligns with observations from the Dhidhessa River Basin, where Dawit et al. [13] found that irrigation potential was highest in areas with slopes of less than 8%, which covered approximately 70% of the basin. These spatial patterns underscore the importance of prioritizing riverine and lowland areas for irrigation investment.

4.1.1.2 Soil suitability Six soil types were identified in the watershed (Table 5 and Fig. 5), with textures ranging from loam to clay. Humic Nitisols and Eutric Vertisols (clay texture, > 120 cm depth, moderately well-drained) were highly suitable, covering 30.67% of the area (975.76 km²). Lithic Leptosols, limited by shallow depth (< 30 cm), were classified as not suitable (25.80% of area), while Eutric Leptosols and Luvisols

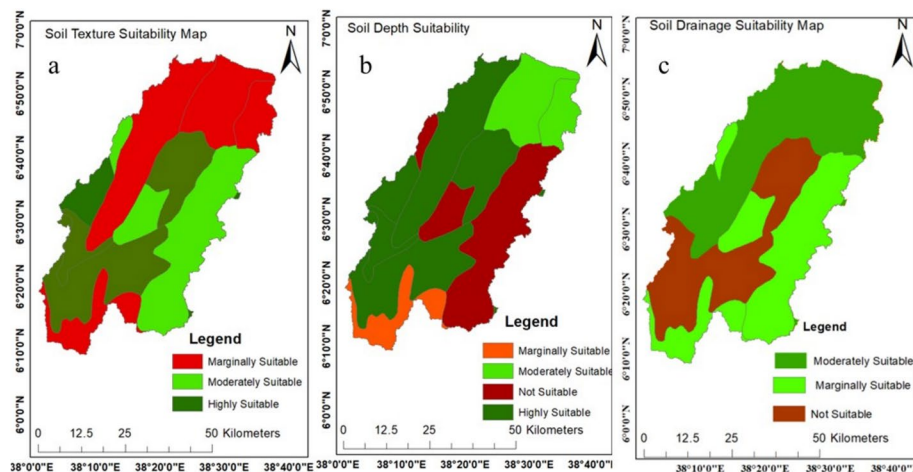


Fig. 5 Suitability map of soil texture (a), soil depth (b), and soil drainage (c)

showed marginal suitability due to texture limitations. The predominance of Humic Nitisols and Eutric Vertisols in the valley bottoms and lowland areas corresponds with findings from the Becho Plain, where Gonfa et al. [25] reported that Vertisols and Nitisols were the most favorable soil types for irrigation, covering approximately 36% of the study area. These soils are characterized by deep profiles and high water-holding capacity, making them particularly suitable for irrigated agriculture. In the Dhidhessa River Basin, Dawit et al. [13] found that soil properties, particularly soil depth and saturated hydraulic conductivity, were among the most sensitive parameters in SWAT model simulations, underscoring their critical role in determining irrigation suitability. The presence of Lithic Leptosols on steep slopes (25.80% of the watershed) creates a compounded constraint, effectively excluding large portions of the upper watershed from irrigation consideration, similar to observations in the Gololcha watershed where shallow soils were associated with steep terrain [2].

Soil depth analysis (Table 5 and Fig. 5) indicated that 50.27% of the watershed (1,598.93 km²) has soil depth > 120 cm (highly suitable), concentrated in valley bottoms and lowland areas. Shallow soils (< 30 cm) covering 25.80% (820.46 km²) were restricted to steep slopes and ridge tops, predominantly associated with Lithic Leptosols. This distribution mirrors findings from the Dendi District, where Legesse et al. [29] reported that approximately 78% of the area had soil depths exceeding 100 cm, classified as highly suitable for surface irrigation. The relationship between soil depth and topographic position is consistent with observations from the Upper Awash Basin, where deeper soils were found in depositional areas with gentle slopes [1].

Soil drainage assessment (Table 5 and Fig. 5) revealed that 40.68% of the watershed (1,293.71 km²) has moderately well-drained soils, primarily in mid-slope positions. Poorly drained soils (25.66%, 816.02 km²), associated with Vertisols in low-lying areas, require careful drainage management, while imperfectly drained soils occupied 33.67% (1,070.76 km²). While these clay-rich soils have high water-holding capacity and are productive under proper management, they are susceptible to waterlogging if surface irrigation is not accompanied by adequate drainage infrastructure. In surface irrigation systems, water ponding is typically caused by operational factors, such as poor

land leveling, excessive inflow rates, or inadequate furrow design, rather than by soil texture alone. However, Vertisols present specific challenges, including slow infiltration rates and swelling characteristics that increase the risk of surface sealing and runoff if not properly managed. This drainage pattern is similar to that reported by Birhanu et al. [11] in the Dirma River Basin, where 51.5% of the area was well drained, and 48.5% was imperfectly drained. The occurrence of poorly drained Vertisols in lowland areas presents both opportunities and challenges: while these soils have high water-holding capacity, they require careful drainage management to prevent waterlogging and salinity accumulation.

4.1.1.3 The overall soil suitability The results of soil suitability in the weighted overlay analysis revealed that about 5.83%, 59.09%, 26.74%, and 8.34% of soils in the watershed were categorized under highly, moderately, marginally suitable, and not suitable classes, respectively, based on the combined effect of the three rasterized soil (drainage, depth, and texture) suitability classes (Table 6; Fig. 6). Based on an integrated analysis of texture, depth, and drainage, 64.92% of the area (S1 and S2) was classified as suitable for surface irrigation, while the remaining 35.08% was not suitable for surface irrigation development.

4.1.1.4 Land use/land cover suitability Nine LULC classes were identified (Fig. 3) and validated using a confusion matrix, achieving an overall accuracy of 88.75% and a kappa coefficient of 0.8725, indicating strong agreement between classified and reference data [7, 15]. Agricultural land (479.63 km², 15.10%) is classified as highly suitable, and grassland (734.36 km², 23.09%) is classified as moderately suitable for irrigation conversion (Fig. 7 and Table 7). Bushland and mixed forest (1,006.99 km², 31.66%) were marginally suitable, whereas settlements, barren land, water bodies, evergreen forest, and wetlands (30.17%) were classified as unsuitable for irrigation development.

The classification of bushland and mixed forest as marginally suitable reflects the need for land clearance investments, similar to observations in the Gololcha watershed, where dense shrubland accounted for 26.97% of the area and was considered marginally suitable Abagissa et al. [2]. Excluding forested areas, settlements, and water bodies from irrigation considerations aligns with environmental protection concerns and infrastructure limitations noted in previous studies [16, 20].

4.1.1.5 Proximity to rivers and roads Euclidean distance analysis showed that approximately 35% of the watershed lies within 1.5 km of perennial rivers (highly suitable), while 52% lies within 3 km of roads (highly suitable), indicating favorable accessibility for irri-

Table 6 Soil suitability of surface irrigation in the weighted overlay analysis

S/N	Area (km ²)	% of area	Suitability
1	185.34	5.83	Highly suitable
2	1879.35	59.09	Moderately suitable
3	850.46	26.74	Marginally suitable
4	265.34	8.34	Not suitable
Total	3180.49	100	

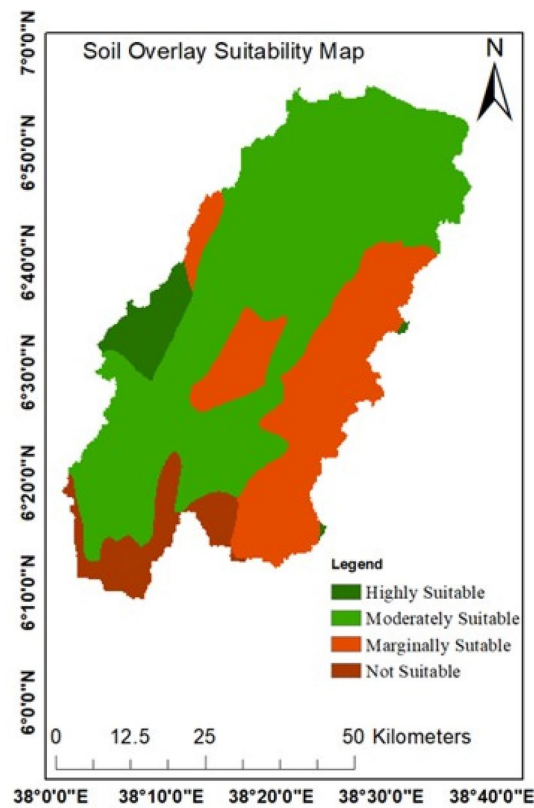


Fig. 6 Integrated soil suitability map for surface irrigation in the Gidabo Watershed, combining soil texture, depth, and drainage parameters through weighted overlay analysis

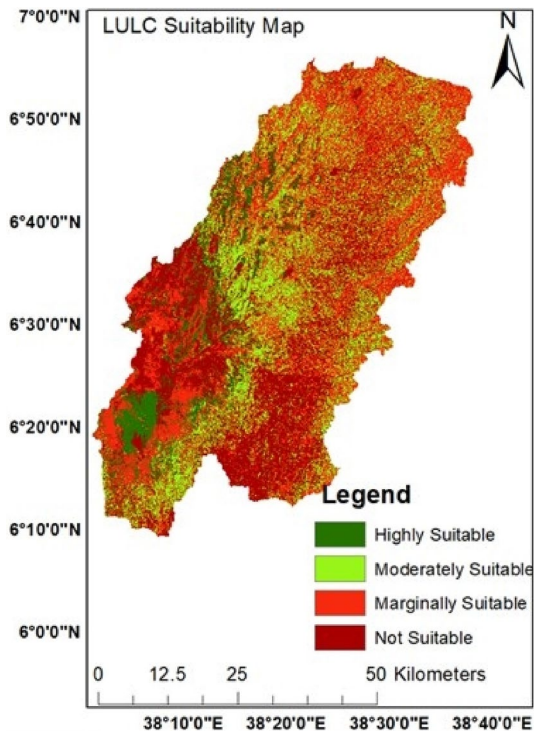


Fig. 7 Land use/land cover suitability classification for surface irrigation in the Gidabo Watershed

Table 7 LULC suitability class

No	Parameters	Suitability			
		Highly suitable%	Moderately suitable%	Marginally suitable%	Not suitable%
1	LULC	15.1	23.09	31.66	33.17

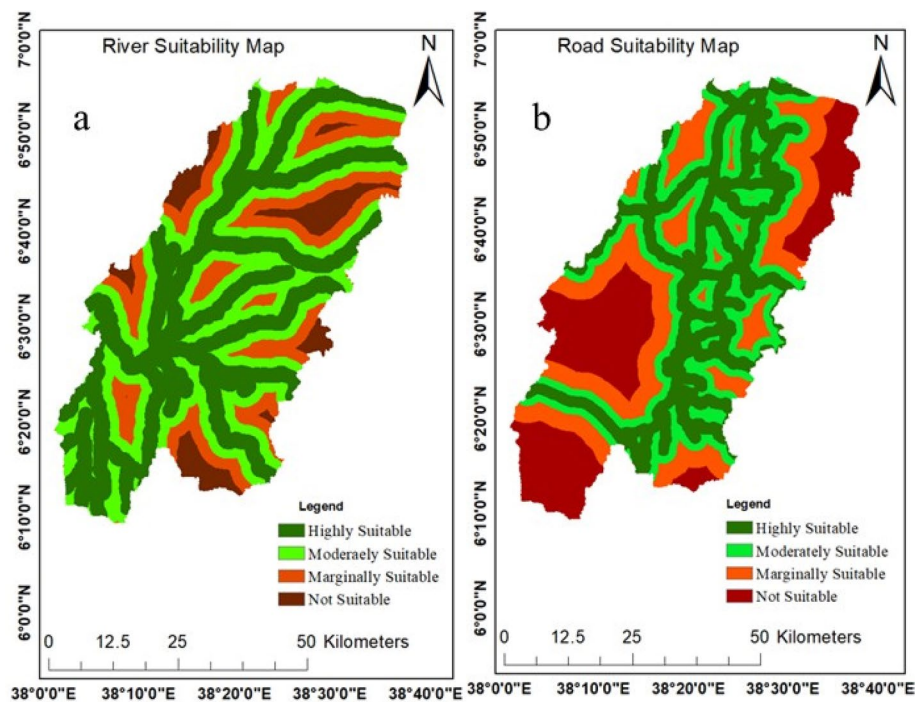


Fig. 8 Map showing the suitability of the study area based on the Distance to rivers (a) and distance to the road (b)

gation development in lowland areas. Areas beyond 5 km from rivers (28%) and 10 km from roads (12%) were concentrated in remote highland regions (Fig. 8). The spatial distribution of river proximity in the Gidabo Watershed, with suitable areas concentrated along major tributaries, mirrors patterns observed in the Dhidhessa River Basin, where irrigation potential was highest in downstream reaches with better water access [13]. The concentration of road access in lowland areas of the Gidabo Watershed is consistent with the spatial pattern of infrastructure development in the Ethiopian Rift Valley, where transportation networks are more developed in accessible lowland areas.

4.1.2 AHP weighting and consistency

The relative importance of the five parameters, as determined by expert judgment through pairwise comparison, is presented in Table 8. The normalized priority weights derived from the eigenvector method (Table 9) identified slope as the most critical factor influencing surface irrigation suitability in the Gidabo watershed, with a weight of 46.36%. This was followed by river proximity (28.51%), soil (14.46%), road proximity (6.01%), and LULC (4.64%) (Table 9). The dominance of slope as the most influential factor (46.34%) is consistent with FAO land evaluation guidelines and numerous Ethiopian

Table 8 Pair-wise comparison matrix

Factor	Soil	Factor LULC	River	Road	Slope
Soil	1	4	1/3	4	1/5
LULC	1/4	1	1/5	1/2	1/7
River proximity	3	5	1	6	1/2
Road proximity	1/4	2	1/6	1	1/7
Slope	5	7	2	7	1

Table 9 Normalized pair-wise comparison matrix

Factors	Soil	LULC	River proximity	Road proximity	Slope	Weight	%
Soil	0.105263	0.210526	0.090090	0.216216	0.100705	0.144560	14.46
LULC	0.026316	0.052632	0.054054	0.027027	0.071954	0.046397	4.64
River proximity	0.315789	0.263158	0.270270	0.324324	0.251762	0.285061	28.51
Road proximity	0.026316	0.105263	0.045045	0.054054	0.071954	0.060526	6.05
Slope	0.526316	0.368421	0.540541	0.378378	0.503525	0.463436	46.34
Sum	1	1	1	1	1	1	100

Table 10 Pair-wise comparison matrix output generated by AHP

Layer name	Soil	LULC	River proximity	Road proximity	Slope	Weight %	CI	RI	CR
Soil	1	4	0.3333	4	0.2	14.46	0.058933	1.12	0.052619
LULC	0.25	1	0.2	0.5	0.1428	4.64	0.058933	1.12	0.052619
River proximity	3	5	1	6	0.5	28.51	0.058933	1.12	0.052619
Road proximity	0.25	2	0.1667	1	0.1428	6.05	0.058933	1.12	0.052619
Slope	5	7	2	7	1	46.34	0.058933	1.12	0.052619
Average CR									0.052619

irrigation suitability studies [4, 19]. The high weight assigned to river proximity (28.51%) underscores the importance of water source accessibility for surface irrigation systems, similar to findings from the Dendi District where proximity to rivers was weighted at 40.74% [29].

The consistency ratio (CR=0.053) (Table 10) was below the 0.10 threshold, confirming acceptable judgment consistency. The consistency ratio of 0.053 is well below the maximum allowable threshold of 0.10 recommended by Saaty [35], indicating that the pairwise comparisons were logically consistent.

4.1.3 Integrated irrigation suitability

The integrated GIS–AHP analysis revealed that approximately 93.23% of the Gidabo watershed is physically suitable for surface irrigation (S1–S3) when only biophysical factors are considered (Fig. 9 and Table 11). However, only 35.79% of the watershed falls within the highly to moderately suitable classes (S1–S2), which are typically regarded as the most feasible areas for irrigation development. This result highlights the substantial spatial variability in irrigation suitability across the watershed,

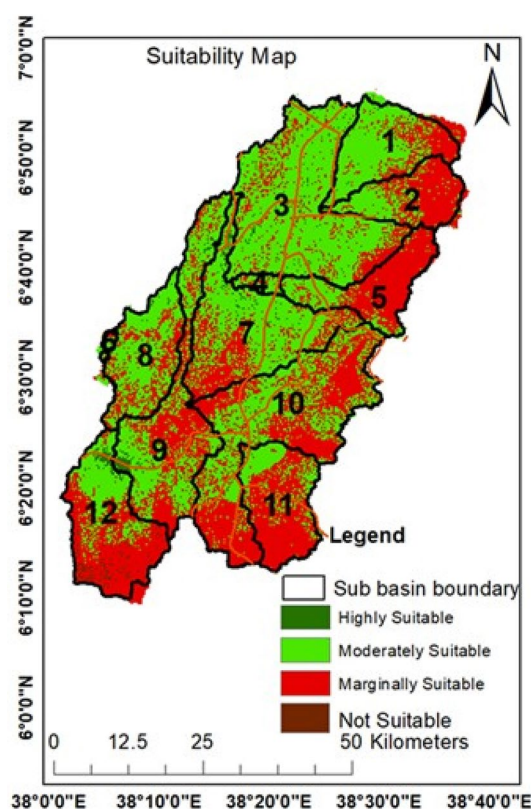


Fig. 9 Final integrated irrigation suitability map of the Gidabo Watershed, showing four suitability classes (S1: Highly suitable, S2: Moderately suitable, S3: Marginally suitable, N: Not suitable) based on GIS-AHP analysis of biophysical and infrastructure factors

Table 11 Integrated land suitability for surface irrigation

Suitability class	Area (ha)	Area (km ²)	Area (%)
Highly suitable (S1)	13,731.85	137.32	4.32
Moderately suitable (S2)	100,099.04	1,000.99	31.47
Marginally suitable (S3)	182,699.67	1,827.00	57.44
Not suitable (N)	21,518.63	215.19	6.77
Total	318,049.00	3,180.49	100

reflecting the combined influence of topography, soil characteristics, land use, and accessibility. Similar findings have been reported in several Ethiopian basins, where large portions of land are physically suitable for irrigation but vary widely in the degree of suitability due to terrain complexity and soil limitations [14, 24, 26, 32].

Topography, particularly slope, emerged as the most influential factor in determining irrigation suitability, accounting for 46.34% of the total weight in the AHP analysis. This dominance of slope as a determining factor is consistent with FAO land evaluation guidelines and numerous irrigation suitability studies [4, 19]. Surface irrigation systems require gentle gradients to ensure uniform water distribution, minimize erosion risk, and reduce land preparation costs. In the Gidabo watershed,

steep slopes (> 8%) cover over 62% of the total area, particularly in the highland zones associated with the Soka Sonicha mountain range. These steep terrains significantly constrain irrigation development because they require extensive land leveling, terracing, or engineered conveyance systems, which substantially increase capital costs and operational complexity. Studies in similar highland watersheds of Ethiopia have reported that land preparation costs may increase by 30–60% when slopes exceed 5%, making irrigation investments economically challenging for smallholder farmers [9]. Conversely, lowland plains and valley bottoms along the Gidabo River and its tributaries exhibit slopes below 5%, which are ideal for surface irrigation development. These areas closely correspond to the highly and moderately suitable classes identified in the suitability analysis, suggesting that irrigation expansion should prioritize these geomorphological settings. Similar spatial patterns have been observed in other Ethiopian basins, such as the Awash and Abbay basins, where irrigation development is concentrated in alluvial plains and river valleys characterized by gentle slopes and deeper soils [16, 28].

Soil properties represent another key determinant of irrigation suitability in the watershed. Approximately 64.9% of the soils were classified as suitable for irrigation, with Humic Nitisols and Eutric Vertisols identified as the most favorable soil types. These soils are characterized by deep profiles (> 120 cm), high clay content, and high water-holding capacity, making them particularly suitable for irrigated agriculture. Vertisols, in particular, are known for their capacity to retain large amounts of moisture, reducing irrigation frequency and improving drought resilience. However, Lithic Leptosols, which cover approximately 25.8% of the watershed, present significant limitations due to shallow soil depth and low water retention capacity. These soils are typically associated with steep slopes and rocky outcrops in highland areas, limiting their suitability for irrigation development. The coexistence of shallow soils and steep slopes creates a compounded constraint, effectively excluding large portions of the upper watershed from irrigation consideration.

Land-use/land-cover patterns also influence irrigation feasibility. Agricultural land and grassland collectively account for approximately 38% of the watershed, representing areas that could potentially be converted to irrigated agriculture with relatively minimal environmental disruption. In contrast, forested and settlement areas were classified as unsuitable due to environmental protection concerns and infrastructure limitations. Similar patterns have been reported in GIS-based irrigation suitability studies conducted in Ethiopia and other developing regions, where land-use constraints often reduce the land available for irrigation [16, 20, 23, 24].

Infrastructure accessibility further refines irrigation suitability. Proximity to rivers was identified as the second most influential factor (28.51%), underscoring the importance of water-source accessibility for surface irrigation systems. Areas located within 2 km of perennial rivers were considered highly suitable due to lower water conveyance costs and reduced infrastructure requirements. Conversely, areas located far from water sources require longer canals or pumping systems, increasing both investment and operational costs.

Overall, the land suitability results demonstrate that while a large proportion of the watershed is technically suitable for irrigation, topography and soil limitations

significantly reduce the area that can be economically developed. These findings emphasize the importance of integrating multiple criteria when assessing irrigation potential, rather than relying solely on individual physical parameters.

4.2 Potentially irrigable lands in each sub-watershed

Sub-basin-level analysis revealed substantial variability in the distribution of suitable areas (Table 12). Sub-basin 3 contained the largest suitable area (214.86 km²), followed by sub-basin 12 (154.98 km²) and sub-basin 10 (146.74 km²). Sub-basin 6 had the smallest suitable area (4.71 km²), consistent with its steep topography and shallow soils.

The spatial variability in suitable land distribution across sub-basins reflects the heterogeneous topography and soil patterns of the Gidabo Watershed. Sub-basin 3, with the largest suitable area (214.86 km²), is characterized by extensive valley bottoms and alluvial plains along the main river channel, similar to the most productive sub-basins in the Dhidhessa River Basin, where Dawit et al. [13] found that downstream sub-basins had the highest irrigation potential. The extremely limited suitable area in sub-basin 6 (4.71 km²) is comparable to findings from the Jewuha watershed, where some sub-basins had minimal irrigation potential due to steep slopes and shallow soils [16].

The percentage of sub-basin area classified as suitable ranges from 5.6% (sub-basin 11) to 57.4% (sub-basin 6). Notably, sub-basin 6, despite having the smallest absolute suitable area, has the highest percentage of its area suitable (57.4%), indicating that where suitable land exists in this sub-basin, it is highly concentrated. This pattern is similar to that observed in the Keleta watershed, where upper catchments had limited but concentrated suitable areas [1]. The low percentage in sub-basin 11 (5.6%) reflects severe topographic and soil constraints, underscoring the importance of sub-basin-scale planning for irrigation development.

Table 12 Irrigable suitable area by sub-basin

Sub-basin	Total area (ha)	Irrigable area (ha) (S1 and S2)	Irrigable area (%)	Rank
Sub-basin 1	20,761.69	9,703.02	46.7	7
Sub-basin 2	22,702.09	6,653.89	29.3	8
Sub-basin 3	54,978.20	21,485.90	39.1	1
Sub-basin 4	2,191.70	1,160.65	53.0	11
Sub-basin 5	18,012.00	5,290.70	29.4	9
Sub-basin 6	820.61	471.34	57.4	12
Sub-basin 7	36,463.00	12,865.00	35.3	5
Sub-basin 8	21,357.90	10,604.14	49.7	6
Sub-basin 9	37,354.00	14,226.25	38.1	4
Sub-basin 10	53,374.00	14,674.00	27.5	3
Sub-basin 11	21,498.00	1,198.00	5.6	10
Sub-basin 12	28,536.00	15,498.00	54.3	2
Total	318,049.19	113,830.90	35.8	

5 Discussion

5.1 Dominant controls on irrigation suitability

The AHP-derived weights identified slope (46.34%) as the most influential factor determining surface irrigation suitability in the Gidabo Watershed, followed by proximity to rivers (28.51%), soil properties (14.46%), proximity to roads (6.01%), and land use/land cover (4.64%). The primacy of slope aligns with established land evaluation principles and numerous Ethiopian irrigation suitability studies [4, 25]. Surface irrigation systems require gentle gradients to ensure uniform water distribution, minimize the risk of erosion, and reduce land preparation costs. The finding that 62% of the watershed has slopes exceeding 8% represents the most significant topographic constraint, effectively excluding the highland escarpments of the Soka Sonicha mountain range from consideration for irrigation. This proportion is comparable to findings from the Dendi District, where Legesse et al. [29] reported 54.38% of slopes as unsuitable, and the Gololcha River Watershed, where Abagissa et al. [2] found 36% of slopes unsuitable due to slope constraints. The slightly higher proportion in the Gidabo Watershed reflects its steeper overall topography, characteristic of the Rift Valley escarpment zone.

The high weight assigned to proximity to rivers (28.51%) underscores the critical importance of access to water sources for surface irrigation systems. This finding is consistent with the Dendi District, where Legesse et al. [29] assigned river proximity a weight of 40.74%, and the Becho Plain, where Gonfa et al. [25] identified water-source proximity as a primary determinant. The concentration of suitable areas within 2 km of perennial rivers (35% of the watershed) reflects the practical reality that conveyance costs increase substantially with distance from water sources. Areas more than 4 km from rivers (28% of the watershed) would require pumping or extended canal systems, increasing both capital costs and operational energy requirements and making them economically unviable for smallholder irrigation schemes [9].

Soil properties, though weighted lower than slope and river proximity (14.46%), remain critically important. The integrated soil suitability analysis classified 64.92% of soils as suitable (S1–S2) for irrigation, with Humic Nitosols and Eutric Vertisols being the most favorable. These soils are characterized by deep profiles (> 120 cm), high clay content, and high water-holding capacity (150–200 mm/m), making them particularly suitable for irrigated agriculture. The 50.27% of the watershed with soil depth >120 cm represents a major comparative advantage for irrigation expansion, as deep soils enhance root proliferation, moisture storage, and nutrient buffering [4]. However, the presence of Lithic Leptosols on steep slopes (25.80% of the watershed) creates a compounded constraint, effectively excluding large portions of the upper watershed from consideration for irrigation. This co-occurrence of shallow soils and steep terrain mirrors observations from the Gololcha watershed, where shallow soils were consistently associated with steep slopes [2], and from the Upper Awash Basin, where deeper soils were found in depositional areas with gentle slopes [1].

5.2 Spatial patterns and environmental gradients

The integrated land suitability analysis revealed a pronounced spatial gradient in irrigation suitability across the Gidabo Watershed. Highly suitable lands (S1, 4.32%) are

concentrated in valley bottoms and lowland plains along the Gidabo River and its major tributaries, where gentle slopes ($<2\%$), deep soils (>120 cm), and favorable drainage conditions converge. Moderately suitable lands (S2, 31.47%) occupy adjacent areas with slopes of 2–5% and moderately well-drained soils, while marginally suitable lands (S3, 57.44%) dominate the mid-slope positions where slope (5–8%) and imperfect drainage impose significant limitations. Not suitable lands (N, 6.77%) are restricted to steep slopes ($>8\%$), shallow soils, and protected land uses (evergreen forest, wetlands, built-up areas).

This spatial pattern corresponds to the physiographic transition from the Gamo Highlands to the Rift Valley floor. The northern upstream sectors feature gentle slopes, deep Nitisols, and near-neutral pH, creating optimal edaphic conditions for surface irrigation. In contrast, the southern and eastern zones exhibit steeper slopes, shallower Leptosols, and more acidic conditions, which limit the feasibility of intensive surface irrigation without significant amendment. This gradient is consistent with observations from the Dhidhessa River Basin, where Dawit et al. [13] found that irrigation potential was highest in downstream reaches with gentle slopes and deeper soils, and from the Becho Plain, where Gonfa et al. [25] reported that Vertisols and Nitisols in valley bottoms were the most favorable soil types.

The proportion of highly suitable land (4.32%) in the Gidabo Watershed is comparable to, but slightly lower than, that reported in other Ethiopian basin-scale studies. For example, Girma et al. [24] reported 8.5% S1 land in the Omo-Gibe Basin, while Muluneh et al. [34] identified approximately 6% highly suitable land across the Rift Valley Lakes Basin. The lower proportion in the Gidabo watershed reflects its steeper overall topography and the more restrictive classification thresholds applied in this study, particularly the reduction of the S2 river proximity upper bound from 4 to 3 km based on local expert consensus (10/12 agreement). This adjustment acknowledges the practical constraints of canal construction in the dissected topography of the Gidabo River valley, where conveyance losses increase substantially beyond 3 km.

5.3 Comparison with regional assessments

The finding that 93.23% of the watershed is physically suitable when marginally suitable lands are included (S1–S3) aligns with the narrative of substantial irrigation potential in Ethiopian basins. However, the critical insight is that only 35.79% falls within the highly to moderately suitable classes (S1–S2), which are economically viable for development. This distinction between physical suitability (technical feasibility) and economic suitability (cost-effectiveness) is often blurred in irrigation planning, leading to overestimation of the developable area [1, 14].

The S1–S2 proportion of 35.79% (113,831 ha) is consistent with the Dendi District, where Legesse et al. [29] reported 35.8% of the area as S1–S2, and the Gololcha watershed, where Abagissa et al. [2] found 40% S1–S2. This convergence suggests that, despite variations in local topography, approximately one-third of Ethiopian highland watershed areas may be economically suitable for surface irrigation when biophysical and infrastructure factors are systematically integrated. The remaining two-thirds are constrained by either steep slopes, shallow soils, protected land uses, or excessive distance from water sources or roads.

The sub-basin-level analysis revealed substantial heterogeneity in the distribution of suitable land, with S1–S2 areas ranging from 471 ha (sub-basin 6) to 21,486 ha (sub-basin 3). This variability has important implications for investment prioritization. Sub-basins 3, 12, and 10 together account for 51.7% of the total suitable area (S1–S2), indicating that targeted investments in these sub-basins could achieve significant irrigation expansion with minimal infrastructure costs. Conversely, sub-basins 6, 11, and 4 have limited suitable area (< 1,200 ha each), suggesting that large-scale irrigation development is not feasible in these areas, although small-scale, community-managed schemes may still be viable [30].

5.4 Implications for irrigation planning and policy

The land suitability results have direct implications for irrigation development strategies in the Gidabo Watershed and analogous Ethiopian highland basins. First, the dominance of slope as a limiting factor suggests that surface irrigation should be restricted to areas with slopes < 5% (S1–S2). Areas with slopes of 5–8% (S3) may be considered for irrigation only with substantial terracing investments, which increase capital costs by an estimated 30–60%, according to regional studies [9]. Given the prevalence of smallholder farming systems with limited access to credit, S3 lands should be deferred for irrigation development until a cost–benefit analysis is completed or reserved for alternative uses such as rain-fed agriculture, agroforestry, or grazing.

Second, the high weight assigned to river proximity (28.51%) supports prioritizing areas within 2 km of perennial rivers for immediate irrigation investment. These areas minimize conveyance costs and reduce the need for pumping infrastructure, making them viable for smallholder schemes. Areas within 2–3 km of rivers (S2) remain viable but may require public subsidy or credit support. Areas more than 3 km from rivers should be excluded from surface irrigation planning unless conjunctive use of ground-water or water storage infrastructure is feasible [37].

Third, the soil suitability analysis provides crop-specific guidance. 64.9% of soils are classified as suitable (S1–S2) and are dominated by Humic Nitosols and Eutric Vertisols, which have high water-holding capacity (150–200 mm/m), allowing longer irrigation intervals (7–10 days) and reducing labor requirements. These soils are highly suitable for water-intensive crops such as maize and tomato. However, Vertisols in low-lying areas (25.66% of the watershed) have high clay content and are classified as poorly drained under natural conditions. These soils require careful water management, including land leveling, controlled furrow inflow rates, and provision of drainage outlets to prevent waterlogging. In surface irrigation systems, water ponding is primarily a function of inadequate land preparation and excessive discharge rather than soil texture. Nevertheless, the swelling characteristics and slow infiltration rates of Vertisols make them more sensitive to these operational factors, necessitating site-specific design considerations. For these areas, raised bed planting or subsurface drainage should be specified in irrigation system designs, following recommendations from the Becho Plain [25] and the Dhidhessa River Basin [13].

Fourth, the LULC analysis indicates that agricultural land (15.10%) and grassland (23.09%) represent the most suitable categories for irrigation conversion, as they require minimal land clearance and have established access. Converting these areas to irrigated

agriculture may displace existing rain-fed production, requiring careful transition planning to avoid disruptions to food security. Bushland and mixed forest (31.66%, marginally suitable) would require clearance investments, which should be factored into project economics. Evergreen forests, wetlands, and water bodies (classified as not suitable) should be protected from irrigation development to maintain ecosystem services, including dry-season baseflow maintenance, sediment retention, and biodiversity conservation [16, 20].

5.5 Methodological contributions and transferability

This study contributes methodologically by demonstrating a transparent, reproducible, and locally validated approach for land suitability assessment in data-scarce regions. The integration of three key methodological elements enhances the credibility and applicability of the results.

First, the use of a 12-member expert panel from three institutional categories (academic researchers, government water resources specialists, and agricultural experts) for factor validation and threshold adjustment ensures that the suitability criteria reflect both international standards (FAO guidelines) and local conditions. The two-round Delphi-style process, with consensus defined as ≥ 10 of 12 experts agreeing, provides a rigorous basis for threshold determination that can be replicated in other basins. The detailed documentation of expert composition, credentials, and voting outcomes (Tables 1 and 2) enhances transparency and allows readers to assess the credibility of the weight assignments.

Second, the AHP weight determination process, conducted by a subset of 10 experts with demonstrated familiarity with comparative judgment methods, yielded a consistency ratio of 0.053, well below the 0.10 threshold. This indicates that the pairwise comparisons were logically consistent and that the derived weights (slope 46.34%, river proximity 28.51%, soil 14.46%, road proximity 6.01%, LULC 4.64%) are reliable for the overlay analysis. The consistency ratio of 0.053 is comparable to or better than those reported in similar Ethiopian studies: Legesse et al. [29] reported CR=0.07, Abagissa et al. [2] reported CR=0.09, and Gonfa et al. [25] reported CR=0.08. This consistency across studies suggests that the AHP method produces stable weight estimates when applied to irrigation suitability assessment in Ethiopian highland contexts.

Third, the sub-basin-level disaggregation of suitability results (Table 12) provides actionable spatial intelligence for investment prioritization. Ranking sub-basins by both absolute suitable area and the percentage of sub-basin area suitable enables decision-makers to distinguish between sub-basins with large absolute potential (e.g., sub-basin 3, 21,486 ha) and those with a high concentration of suitable land (e.g., sub-basin 6, 57.4% of its area). This dual ranking is essential for efficient resource allocation: sub-basins with large absolute suitable area justify investment in central infrastructure (canals, intake structures), while sub-basins with high concentration but small absolute area may be better served by small-scale, distributed schemes.

The methodological framework is readily transferable to other data-sparse regions. The core requirements, SRTM DEM, Landsat imagery, national soil maps, and road/river network data, are widely available across sub-Saharan Africa. The expert panel approach can be adapted to local institutional contexts, and the AHP weight determination can

be conducted using free software. The primary limitation for transferability is the availability of local experts with sufficient familiarity with both irrigation engineering and the study area. In regions where such expertise is limited, the threshold values and weights from this study could serve as initial estimates, subject to local validation through sensitivity analyses.

5.6 Limitations and future research directions

Several limitations of the land suitability assessment should be acknowledged. First, the assessment focused on biophysical and infrastructure factors and did not explicitly incorporate socioeconomic variables such as land tenure, farmer capacity, market accessibility, or institutional constraints. The absence of these factors means that areas identified as physically suitable (e.g., sub-basins 10, 3, and 7) may face significant economic or social barriers to actual development, including fragmented landholdings that complicate collective irrigation schemes, limited farmer access to credit for irrigation infrastructure, or competing water uses not captured in the hydrological model [30, 34]. Consequently, the suitable area of 113,831 ha reported here should be interpreted as a biophysically based upper bound rather than a prediction of actual development outcomes. Future research should integrate socioeconomic indicators into multi-criteria decision-making frameworks, potentially through participatory GIS approaches that combine remote sensing with community-based mapping.

Second, the soil data were obtained from a 1:250,000 scale national map, which may not capture local-scale variability in soil properties relevant to farm-level irrigation planning. This is particularly problematic for soil drainage classification, where local variations in topography and subsurface flow paths can significantly affect the risk of waterlogging. Future research should incorporate higher-resolution soil surveys, including field measurements of saturated hydraulic conductivity and infiltration rates, to refine the suitability classification at the sub-basin scale.

Third, the assumption that proximity to roads (weighted at 6.01%) adequately captures market accessibility may be simplistic, as road quality, seasonal accessibility, and distance to specific markets (rather than any road) influence transaction costs. Future research should incorporate more detailed infrastructure data, including road surface type, all-weather access, and travel time to major market centers, to refine the accessibility component of the suitability assessment.

Fourth, the land suitability assessment did not consider alternative irrigation technologies (e.g., drip or sprinkler) that may be feasible on steeper slopes or poorer soils. While surface irrigation remains the most common and affordable technology for smallholder farmers in Ethiopia, future research should evaluate the suitability of alternative systems for S3 and N lands, comparing their capital costs, water-use efficiency, and crop productivity with surface irrigation on S1–S2 lands.

Finally, we acknowledge that sensitivity analysis could provide additional insight into the robustness of the AHP-derived weights; such analysis was beyond the scope of the current study. The inherent consistency checking within the AHP process ($CR=0.053$), the strong expert consensus (≥ 10 of 12 experts on all thresholds), and the alignment of our weights with comparable studies (e.g., 46.34% slope weight matching Abagissa et al. [2] provide confidence in our results without formal

sensitivity testing. Future research could conduct one-at-a-time (OAT) or global sensitivity analyses to further validate the weight assignments.

6 Conclusion

This study assessed where surface irrigation is most suitable in the Gidabo Watershed by combining GIS, AHP, and information on slope, soils, land use, and access to rivers and roads. The analysis showed that 93.23% of the watershed is physically suitable to some degree (S1–S3). However, only 35.79% (113,831 ha) falls within the highly and moderately suitable classes (S1–S2). This difference is important because it shows that the total area that appears physically suitable can substantially overstate the area with favorable conditions for practical irrigation development. Highly suitable land covers only 4.32% of the watershed and is mainly concentrated in valley bottoms and lowland plains along the Gidabo River and its major tributaries.

Slope was the most important factor in the assessment, with an AHP weight of 46.34%. More than 62% of the watershed has slopes greater than 8%, which represents the main topographic constraint to conventional surface irrigation. Distance to rivers was the second most important factor (28.51%), highlighting the importance of access to reliable water sources and the cost of conveying water to fields. Soil properties also influenced suitability, particularly in areas with shallow soils or poor drainage.

The distribution of suitable land varied considerably among the 12 sub-basins. Sub-basin 3 contained the largest S1–S2 area (21,486 ha), followed by sub-basins 12 and 10, while sub-basin 6 had the smallest area (471 ha). Together, sub-basins 3, 12, and 10 contained 51.7% of the total S1–S2 area. These areas, therefore, provide an important starting point for more detailed irrigation feasibility studies and investment planning. However, the suitability map should be regarded as a screening and planning tool rather than a substitute for site-specific engineering, economic, environmental, and water-availability assessments.

Overall, the study reveals that combining spatial information on terrain, soils, land use, and infrastructure can help identify areas with more favorable conditions for surface irrigation development in a data-scarce watershed. The results can support more targeted field investigations, help guide irrigation investment planning in the Gidabo watershed, and provide a useful basis for similar assessments in other heterogeneous watersheds in sub-Saharan Africa. Future studies should complement the present assessment with detailed soil and field surveys, water-availability analysis, economic feasibility assessment, and sensitivity analysis to further refine irrigation development priorities.

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Author contributions

Henok Mekonnen Aragaw: Methodology, conceptualization, writing, review & editing. Azmeraw Alemu Tiku: Data curation, methodology, investigation, conceptualization, formal analysis & writing.

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Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study protocol was reviewed and deemed exempt by the Dilla University Research Ethics Board. The study was conducted in accordance with Dilla University's educational research policies. All procedures involving human participants, including expert consultations, key informant interviews, and field surveys with local farmers and agricultural extension agents, were performed in compliance with Dilla University's research ethics standards.

Informed consent

Informed consent was obtained from all participants involved in the study. Participants were provided with comprehensive information about the study's purpose, procedures, and data use prior to their participation, and they confirmed their voluntary participation.

Consent for publication

Not applicable.

Competing interests

The authors declare no conflict of interest.

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